3 **(a)** State what is meant by the internal energy of a system.

.....

.....

..... [2]

(b) Use the first law of thermodynamics to explain what happens to the internal energy:

(i) of a spring when it is stretched at constant temperature within its elastic limit

.....

.....

.....

.....

..... [3]

(ii) of a sample of water when it evaporates from a rain puddle on a hot day.

.....

.....

.....

.....

..... [3]

[Total: 8]